MATH 170 Calculus I

Course Information

MATH 170 01
MTWRF 9:00-9:50
Taylor 106

Contact Information

Instructor	Jennifer Nordstrom
Office	Taylor 205
Drop in Hours	MTWR 1:30-3:00 But feel free to drop by any time.
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Course Description

Differential and integral calculus of real functions of one variable. Differentiation, the chain rule, the mean-value theorem, the fundamental theorem, limits and continuity, curve sketching. Integration by substitution. Application of the derivative and integral to physics and geometry.

Course Materials and Links

The text for this course is *Calculus for Team-Based Inquiry*. The text is free and online.

We will also be using the associated Check-It problem bank for quizzes.

We are using WeBWorK for online homework sets. You will need to login using the instructions provided in class.

Dates and assignments for this course are posted on our assignments page.

Links to the course resources are also posted in Blackboard.

Course Goals and Objectives

Objectives: Course Goals

1. Students gain an understanding of the fundamentals of differential and integral calculus and its applications in the physical and social sciences.

2. Students solidify their abilities in the use of algebra, trigonometry, and calculus.

3. Students develop an appreciation for the foundations of the real number system
4. Students increase their problem solving abilities.
5. Students increase their mathematical maturity and confidence in working through mathematical challenges.

The course is based around the following Course Learning Objectives.

1. Find limits graphically, numerically, and analytically. (LT1, LT2, LT3)
2. Determine where a function is and is not continuous. (LT4)
3. Determine limits with infinite inputs and infinite outputs. (LT5, LT6)
4. Find derivatives using the limit definition of the derivative. (DF2)
5. Compute basic derivatives using algebraic rules. (DF3)
6. Compute derivatives using Product, Quotient, and Chain Rules. (DF4, DF5, DF6)
7. Compute derivatives of implicitly-defined functions. (DF7)
8. Compute derivatives of inverse trig functions and log functions. (DF8)
9. Use the derivative to find tangent lines, and use the tangent line for linear approximation. (AD1, AD2)
10. Model and analyze scenarios using related rates. (AD3)
11. Use the Extreme Value Theorem to find global max and min values. (AD4)
12. Use derivatives to determine where a function is increasing, decreasing, concave up, concave down, and to find local extrema and inflection points. (AD5, AD6, AD7)
13. Apply optimization techniques to solve various problems. (AD8)
14. Approximate definite integrals using Riemann sums. (IN2)
15. Determine basic antiderivatives. (IN3)
16. Evaluate a definite integral using the Fundamental Theorem of Calculus. (IN5)
17. Use definite integrals to find the area under a curve and the area between curves. (IN7, IN8)
18. Evaluate various integrals using the substitution method. (TI1)

Daily Schedule

Tentative Daily schedule for Calculus I

Week	Mon	Tues	Wed	Thur	Fri
1	Intro	Review	Review	RAT 1	1.1
2	No class	1.1	1.2	1.2	1.3
3	1.3	1.4	1.4	1.5	Quiz 1
4	1.6	RAT 2	2.1	2.1	2.2
5	2.2	2.3	2.4	2.4	Quiz 2
6	2.5	2.5	2.6	Fall Break	Fall Break
7	2.7	2.8	RAT 3	3.1	Quiz 3
8	3.1	3.2	3.2	3.3	3.3
9	3.4	3.4	3.5	3.6	Quiz 4
10	3.7	3.8	3.8	3.8	RAT 4
11	4.1	4.2	4.3	4.4	Quiz 5
12	4.5	4.6	4.7	4.8	4.8
13	Break	Break	Break	Break	Break
14	5.1	5.1	5.1	6.1	Quiz 6
15	6.3	6.3	6.3	3.9	3.9
16	Reading Day		Final Exam Time 8am		

The Importance of Community

Many aspects of this course are structured to build community. Although learning mathematics is often seen as an individual task, learning is more effective when you engage with others during the process. As a class we will determine the particular attributes we expect of each other in order to build our community. In addition, learning to work well in a community in which members bring different skills will better prepare you for life and work after college. Some ways you can help build a strong supportive community that improves learning are

- Come to class ready to participate.
- Help your team in a variety of ways: volunteer to write on the board, keep and organize notes for your group, ask questions, make suggestions, be positive and encouraging of others.
- Come to office hours for help rather than relying on AI or other tools when you are unsure of the concepts. Your professor is part of the community, too. Building connections with faculty will help you not just learn Discrete Math, but also help connect you with other resources and opportunities.
- Form study groups to work on homework together or review for quizzes. If you use AI to help you study, it is still better to do this in a group so you catch errors.
- Don't be afraid to make mistakes. We all learn from each others' mistakes and misunderstandings. You are not expected to come to this class as a fully-formed mathematician. In fact, you aren't expected to leave this

class as a fully-formed mathematician either. But if no one knows what mistakes you are making, no one can help you learn.

Team-Based Inquiry Learning

This course is taught by a method called **team-based learning**. You will be assigned a team that you will work with on various activities in class each day.

- Before each module begins, you will be responsible for ensuring your own readiness for the module. Material for the readiness assurance process will be available on Blackboard. Make sure you prepare for the beginning of the module before class.
- The first day of each module will be dedicated to the Readiness Assurance Process. The dates for these are located in the Daily Schedule (listed as RAT). On these days, you will first take an Individual Readiness Assurance Check (iRAT). After submitting this, working with your teammates you will retake the same problems as the Team Readiness Assurance Check (tRAT). These are designed to measure if you are prepared for the team activities on subsequent days. See Readiness Assurance Checks for grading information.
- On the other class days, you will work with your teammates on a series of activities designed to guide you through discovering and understanding the course material.

Grading Information

The following includes descriptions of the graded components of the course and the grading scale.

Team and Community

You will be working in teams of 4-6 students on in-class activities. You will not be assessed on your team's specific answers to the activities, but you will be assessed on your general engagement with your team. This includes attendance, participation, and teamwork. Your participation and team-work with your group will count toward your community grade. If you are unable to be in class, it is your responsibility to check with your team about what activities you missed.

At the end of each non-quiz week there will be a short community activity. These will generally be graded for completeness. There will also be a final reflective assignment for you to summarize your overall contributions to our community.

Over-reliance on AI tools to do mathematics is often a replacement for doing mathematics with other people. This can impact your ability to learn, evaluate, and communicate mathematics. Evidence of such use will affect your community grade. See Use of AI in Student-Generated Work for the specifics of appropriate uses for this course.

Readiness Assurance Checks

At the start of each module, you will take a multiple choice assessment (iRAT) based on assigned material. After submitting you answers, you will work with your team on the same assessment (tRAT), submitting team answers. Your scores on both the iRAT and the tRAT will be averaged for your RAT score. If you are unable to be at class for the RAT process, you may take the iRAT within two class days. Your iRAT will stand as your score. If you miss a RAT and are unable to make it up within the two days, I will drop one RAT score.

RAT 1	Thursday, Sep. 3.
RAT 2	Tuesday, Sep. 22.
RAT 3	Wednesday, Oct. 14.
RAT 4	Friday, Nov. 6.

- When? Before each new chapter.
- What if I miss a RAT? You can take the iRAT before the next class, but not the tRAT. Your iRAT will be used for both parts of your RAT score.
- How do I retake a RAT? Schedule a time with me to take it within two days.

Homework

Homework will be due two to three times per week. It will be due at ***11:59 pm*** on the due date. All homework assignments are on WebWork.

In addition to the WeBWorK problems, you are encouraged to do CheckIt problems to practice for the Quizzes: CheckIt for Calc I

I expect you to work on this course outside of class daily. It is highly recommended that you work on the homework throughout the week so you have plenty of opportunity to get help. You can request an extension of any homework assignment. All extensions will be for 48 hours from the original due date and time.

- How to turn in? Submitted through WeBWork.
- Late submission accepted? Yes. You may request a 48 hour extension.

Daily Questions

At the start of class almost every day you will be asked to answer (and turn in your answer to) the Daily Question. These will be questions on previous material. The purpose of these is primarily diagnostic, not evaluative. They will be graded on a pass/ no pass (P/NP) basis. You may receive a NP on five Daily Questions with no penalty, after that, you will lose 3% of the Daily Question grade for each NP. There are absolutely no make-up questions. A missed Daily Question counts as a NP. It is important to arrive to class on time, as these will only take a few minutes. Excused absences such as for athletics or field trips will not count against your Daily Question grade.

- How to turn in? Done on paper and turned in during class.
- Late submission accepted? No.

Proficiency-Based Quizzes

This course contains 18 essential types of problems. The 18 standards are listed in the Course Learning Objectives in Course Goals and Objectives. Each objective is related to the learning outcomes in the text, such as LT1 or DF2, These outcomes also correspond to the Learning Outcomes in the CheckIt bank at the end of each section or at Calc I CheckIt.

Every two weeks there is a quiz. Each quiz will introduce 2 to 4 new objectives, as well as variations on the previous problems. The final quiz day, during the scheduled Final Exam Time, there will only be variations on previous problems.

On each objective you will be scored Proficient (P), Developing (D), or Novice (N). "Proficient" means you have demonstrated a solid mastery of the concept. "Developing" means you are on your way to mastery, but still need to clarify some details. "Novice" means you have yet to demonstrate an understanding of the concept and need to carefully study it some more. Only the score of "Proficient" earns credit for the concept. You may attempt the concepts that you have not earned "Proficient" on again at the next quiz. There are 18 objectives and 6 quizzes, so it is reasonable to study to pass 3 objectives at each quiz. once you have demonstrated proficiency, you do not need to attempt it on a future quiz.

Your quiz grade will be determined by the number of concepts for which you have earned "Proficient." You will get 8 points for each of the first 9 concepts you master, then for the next 9, you will earn 3 points for each "Proficient."

Number of Concepts	Quiz Grade
1	8
2	18
3	24
4	32
5	40
6	48
7	56
8	64
9	72

Number of Concepts	Quiz Grade
10	76
11	79
12	82
13	85
14	88
15	91
16	94
17	97
18	100

Some advantages to this type of grading is that you get several attempts to try to master the material. It is OK to not be able to solve problems on a particular quiz day, but you do need to be able to solve the problems by the end of the course. If there are some concepts that you struggle with more than others (which there will be!) you can always come back to them later. Much of the material is cumulative, so it will be important to keep up with many of the concepts. The main disadvantage is that the system makes it trickier for you to know exactly where you stand, since you can't earn an A or B on you quiz score until we are towards the end of the semester. However, if you need help determining how you are doing in the course, just ask.

Quiz 1	Friday, Sep. 18.
Quiz 2	Friday, Oct. 2.
Quiz 3	Friday, Oct. 16.
Quiz 4	Friday, Oct. 30.

Quiz 5	Friday, Nov. 13.
Quiz 6	Friday, Dec. 4.
Quiz 7	Retakes only. Wednesday, Dec. 16, 8:00 am.

- When? Every other Friday.
- What if I miss a quiz? You can retake any of the standards at the next scheduled quiz time.

Grading System

Grades will be determined as follows:

Community and Team	15%
RATs	10%
Homework	15%
Daily Questions	10%
Proficiency Quizzes	50%
Team and Community	10%

Letter grades correspond to the following percentages:

A-, A	90-100%
B-, B, B+	80-89%
C-, C, C+	65-79%
D	55-64%

Final Exam Time

The final exam time for this class is scheduled for ***Wednesday, December 16 at 8:00 am***.

We will use the final exam time for our last quiz. There will be no new standards. Instead, this will be the final time to retake any previous standards to improve your grade. You do not need to attend the final exam time if you do not wish to retake any previous standards.

Advising Information

The prerequisite for this course is Math 150, Precalculus, which is equivalent to a high school course which includes Trigonometry. This course is designed for anyone who would like to learn calculus. It is required of math, physics, chemistry, and computer science majors. It is also recommended for majors in Economics who intend to pursue graduate study and majors in Biology who intend to pursue graduate or medical school.

Course Policies

Cell Phones

Cell phones must be off and put away during class. Laptop computers and tablets are encouraged, as we will be engaging with materials that are available electronically. However, please use them in ways that are focused on the course and the activities of the class.

Electronic Recording/ Content Sharing

I may opt to record the classroom activities for instructional purposes and post them to the cloud. The electronic recording of classroom lectures, discussions, simulations, and other course-related activity is governed by Linfield's Classroom Recording Policy (Faculty Handbook, VII.26 and Student Policy Guide). Students do not have permission to record any Zoom meetings. Students do not have permission to distribute or share any recorded content from Zoom meetings.

Use of AI in Student-Generated Work

Students in this course are expected to avoid the use AI tools, such as Chat GPT, Gemini, Claude, and PhotoMath, to generate activity, quiz, or homework solutions. Any tools may only be used in a manner that contributes to understanding math, rather than avoiding the work necessary to deepen your understanding. Use of such tools, like any other academic work that is not entirely the student's own, must be cited. Work for a grade that is not primarily in the student's own words and properly cited, will be considered plagiarized. Note, many of the math tools that exist use techniques that are not part of this course.

If a violation is suspected, the first action will be a conversation about why your work may be read as AI generated. Work that is deemed not the student's own after this conversation will receive a 0 on the entire assignment/ activity, in addition to a lower Community grade.

Some AI uses that are appropriate: generating questions to help you study for quizzes, helping summarize key concepts with additional examples, helping create a study plan for balancing your workload. Uses that are not appropriate: generating homework solutions with AI, having AI give you answers for the questions in the Activities, using any AI tool during a quiz or Daily Question.

Linfield Policies

Disability Statement:

Students with disabilities are protected by the Americans with Disabilities Act and Section 504 of the Rehabilitation Act. If you are a student with a disability and feel you may require academic accommodations please contact Learning Support Services (LSS), as early as possible to request accommodation for your disability. The timeliness of your request will allow LSS to promptly arrange the details of your support. LSS is located in Melrose Hall 020 (503-883-2562), or LSS@linfield.edu. We also encourage students to communicate with faculty about their accommodations.

Academic Integrity Policy:

Linfield University operates under the assumption that all students are honest and ethical in the way they conduct their personal and scholastic lives. Academic work is evaluated on the assumption that the work presented is the student's own, unless designated otherwise. Anything less is unacceptable and is considered a violation of academic integrity. Furthermore, a breach of academic integrity will have concrete consequences that may include failing a particular course or even dismissal from the university. Violations of academic integrity include but are not limited to the following:

Cheating: Using or attempting to use unauthorized sources, materials, information, or study aids in any submitted academic work; changing answers after graded work has been returned; making unauthorized changes to an exam, quiz, or assignment.

Plagiarism: Submission of academic work that includes material copied or paraphrased from published or unpublished sources without proper documentation. This includes self-plagiarism, the submission of work created by the student for another class unless they receive consent from both instructors.

Fabrication: Deliberate falsification or invention of any information, data, or citation in academic work.

Facilitating Academic Dishonesty: Knowingly helping or attempting to help another to violate the university's policy on academic integrity.

For specific information about the use of AI, see Use of AI in Student-Generated Work.

Any form of academic dishonesty will result in a 0 on that assignment/ quiz/ exam. Additionally, academic dishonesty may result in a failing grade in the course. See the Linfield Academic Integrity Policy in the Linfield Catalog for information on the procedure to be used in dealing with academic dishonesty.

Sexual Misconduct and Relationship Violence and Title IX:

Linfield University faculty are committed to supporting students and fostering a campus environment free of sexual misconduct and relationship violence. If a student chooses to disclose to a faculty or staff member an experience related to sexual misconduct, sexual assault, domestic violence, dating violence, or stalking, all faculty and staff are obligated to report this disclosure to the Linfield Title IX Coordinator by emailing `titleix@linfield.edu`. Upon receipt of the report, the Title IX Coordinator will contact you to inform you of your rights and options and connect you with support services. If you would rather share information about these experiences with an employee who does not have these reporting responsibilities and can keep the information confidential, please visit confidential resources: linfield.edu/safety/sexual-misconduct/reporting-options/confidential.html.

For more information about your rights and reporting options at Linfield, including confidential reporting options, please visit Linfield's Know Your Options Page on the University website. Support services are offered to all Linfield students regardless of whether or not they report. Still have questions? Email `titleix@linfield.edu`.

Commitment to Diversity and Inclusion:

Linfield University honors human rights and academic freedom, celebrates diverse cultures, fosters a climate of mutual respect, and promotes an inclusive environment that affirms the value of all persons. Dimensions of diversity can include sex, race, age, national origin, immigration status, ethnicity, gender identity and expression, intellectual and physical ability, sexual orientation, income, faith and non-faith perspectives, socio-economic class, political ideology, education, primary language, family status, military experience, cognitive style, and communication style. In a multi-perspective intellectual space, challenges to our beliefs and ideas are part of the learning process and can provide opportunities for growth. Reasoning, thoughtfulness, and open dialogues that honor the dignity of everyone is expected.